

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 4: PANS-OPS Instrument Procedures & Circling

Syllabus Source: CAE Oxford (pp. 117-156) | Pass Target: >= 90% AviationExam

1. PANS-OPS INSTRUMENT DEPARTURE CRITERIA

ICAO Doc 8168 (PANS-OPS) establishes standard criteria for protected airspace design:

- Design Climb Gradient (PDG): Standard departure climb gradient is 3.3% (200 ft/NM), comprising 2.5% net obstacle clearance + 0.8% safety buffer.
- Minimum Turn Altitude: No turn may be initiated on departure below 120 METERS (400 FT) ABOVE AERODROME ELEVATION!

2. VISUAL MANOEUVRING (CIRCLING APPROACH)

A visual circling approach is used when landing runway does not align with the instrument approach (alignment > 30° off runway centerline, or descent gradient exceeds 6.5%):

- Protected Circling Radii: Cat A = 1.68 NM; Cat B = 2.66 NM; Cat C = 4.20 NM; Cat D = 5.28 NM from all runway thresholds.
- Action on Loss of Visual Reference: IF VISUAL CONTACT WITH THE RUNWAY IS LOST DURING CIRCLING: Initiate an IMMEDIATE CLIMBING TURN TOWARDS THE LANDING RUNWAY, establish overhead the aerodrome, and follow the missed approach procedure of the INSTRUMENT RUNWAY originally flown!

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under PANS-OPS design rules, what is the MINIMUM altitude above aerodrome level before an aircraft may

[A] 200 ft [B] 120 METERS (400 FEET) AGL [C] 1,000 ft [D] 500 ft

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS prohibits turns below 120 m (400 ft) above aerodrome elevation to guarantee obstacle clearance during straight acceleration.

Q2: If visual reference is LOST while conducting a visual circling manoeuvre to land, what is the mandatory pil

[A] Descend 500 ft to find the ground [B] Initiate an IMMEDIATE CLIMBING TURN TOWARDS THE LANDING RUNWAY, climb overhead, and fly the missed approach of the instrument approach runway [C] Turn 90° away from airport [D] Land immediately

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. Turning towards the runway keeps the aircraft within the protected circling obstacle area while climbing to safety in the instrument missed approach.

Q3: What is the radius of the protected obstacle clearance area for a CATEGORY C aircraft conducting a circling

[A] 1.68 NM [B] 4.20 NAUTICAL MILES from the runway threshold [C] 2.66 NM [D] 5.28 NM

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS assigns Cat C aircraft a circling area radius of 4.20 NM based on a maximum circling speed of 180 kt.